ECE 310 Fall 2026
Lecture 7: z-transform Properties

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Learning Objectives

- Understand key properties of the z-transform and the region of convergence.
- Apply properties of the z-transform to derive the z-transform of discrete-time signals.



A Closer Look at the z-transform and ROC

We define the z-transform $X(z)$ for discrete-time signal $x[n]$ as

$$X(z) = \sum_{n-\infty}^{\infty} x[n] z^{-n}$$

where we must also denote all values of z where the above summation converges, known as the *region of convergence* (ROC).



A Closer Look at the z-transform and ROC

It is important to note that $X(z)$ is not defined for values of z that lie outside the ROC even if the expression for $X(z)$ computes a finite value.

Consider

$$x[n] = \left(\frac{1}{2}\right)^n u[n] \overset{z}{\longleftrightarrow} X(z) = \frac{1}{1 - \frac{1}{2}z^{-1}}, |z| > \frac{1}{2}$$



A Closer Look at the z-transform and ROC

Let's summarize some important properties for the definition of the z-transform and region of convergence:

- Values of z where $X(z) = 0$ are referred to as *zeros* while values where $X(z) \to \infty$ are *poles*.
- The ROC defines values of z for which the z-transform sum converges. Thus, the ROC cannot include any poles.
- The ROC is a connected region.



A Closer Look at the z-transform and ROC

For finite-length signals, the ROC is the entire complex z-plane, except perhaps $z = 0$ for causal sequences or $z = \infty$ for non-causal sequences.



A Closer Look at the z-transform and ROC

For infinite-length signals, the ROC can have one of the following shapes:

- Right-sided sequence, i.e. $x[n] = 0, n < n_0$: ROC $= \ |z| > a, \ a \geq 0.$
- Left-sided sequence, i.e. $x[n] = 0, n < n_0$: ROC $= \ |z| < a, \ a \geq 0.$
- Two-sided sequence: ROC $= a < |z| < b, \ a \geq 0, b \geq 0, a < b.$



Example 1: Computing ROC

Determine the ROC of the below $X(z)$ assuming the corresponding $x[n]$ is right-sided. What is the ROC if $x[n]$ is left-sided?

$$X(z) = \frac{1 + 3z^{-1}}{1 + \frac{7}{4}z^{-1} - \frac{1}{2}z^{-2}}$$



z-transform Properties: Time-shifting

Let $x[n]$ be a discrete-time signal with z-transform $X(z)$ and ROC R_x.

$$x[n-k] \overset{\mathcal{Z}}{\longleftrightarrow} z^{-k}X(z),\ k \in \mathbb{Z},\ \text{ROC } = R_x \text{ except possibly } z = 0 \text{ or } z = \infty$$



z-transform Properties: Convolution

Let $x_1[n]$ and $x_2[n]$ be discrete-time signals with z-transforms $X_1(z)$ and $X_2(z)$ and ROCs R_{x_1} and R_{x_2}, respectively.

$$x_1[n] * x_2[n] \overset{\mathcal{Z}}{\longleftrightarrow} X_1(z)X_2(z), \ \text{ROC} \ = \ \text{at least} \ R_{x_1} \cap R_{x_2}$$



z-transform Properties: Linearity

Let $x_1[n]$ and $x_2[n]$ be discrete-time signals with z-transforms $X_1(z)$ and $X_2(z)$ and ROCs R_{x_1} and R_{x_2}, respectively.

$$ax_1[n] + bx_2[n] \overset{\mathcal{Z}}{\longleftrightarrow} aX_1(z) + bX_2(z), \text{ ROC } = \text{ at least } R_{x_1} \cap R_{x_2}$$



z-transform Properties: Scaling

Let $x[n]$ be a discrete-time signal with z-transform $X(z)$.

$$a^n x[n] \overset{\mathcal{Z}}{\longleftrightarrow} X\left(\frac{z}{a}\right), \; a \in \mathbb{C}, \; \text{ROC} \; = |a| R_x$$



Example 2: Applying z-transform Properties

Let $x_1[n] = \left(\frac{1}{2}\right)^n u[n] - c\left(\frac{1}{2}\right)^{n-1} u[n-1]$, $x_2[n] = (-1)^n u[n]$ for $c \in \mathbb{R}$, and $y[n] = x_1[n] * x_2[n]$. Determine $Y(z)$ and its ROC when (a) $c = 1$, (b) $c = -1$.



z-transform Properties: Differentiation

Let $x[n]$ be a discrete-time signal with z-transform $X(z)$.

$$nx[n] \overset{\mathcal{Z}}{\longleftrightarrow} -z\frac{dX(z)}{dz}, \ \text{ROC} \ = R_x$$



Example 3: Applying z-transform Properties

Let $x[n] = \left(\frac{1}{3}\right)^n u[n]$. Determine the z-transform $Y(z)$ and ROC of $y[n]$ below.

$$y[n] = 2^n n x[n]$$



Summary of Key z-transform Properties

Property	Signal	z-transform	ROC
Time-shifting	$x[n-k]$	$z^{-k}X(z)$	R_x except $z=0$ or $z=\infty$
Linearity	$ax_1[n]+bx_2[n]$	$aX_1(z)+bX_2(z)$	At least $R_{x_1}\cap R_{x_2}$
Convolution	$x_1[n]*x_2[n]$	$X_1(z)X_2(z)$	At least $R_{x_1}\cap R_{x_2}$
Differentiation	$nx[n]$	$-z\frac{dX(z)}{dz}$	R_x
Conjugation	$x^*[n]$	$X^*(z^*)$	R_x
Time reversal	$x[-n]$	$X\left(z^{-1}\right)$	$1/R_x$
Scaling	$a^n x[n]$	$X\left(z/a\right)$	$\lvert a\rvert R_x$
Real-part	$\mathrm{Re}\{x[n]\}$	$(1/2)[X(z)+X^*(z^*)]$	At least R_x
Imaginary-part	$\mathrm{Im}\{x[n]\}$	$(1/2)[X(z)-X^*(z^*)]$	At least R_x

